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Industrial Internship Report on

”Banking Information System”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on solving a project/problem statement by developing a simple software-based banking application using Java. The project was completed within the internship duration, including coding, testing, and report preparation. My project was “Banking Information System”, which is a console-based Java application designed to perform basic banking operations such as account creation, login, deposit, withdrawal, balance checking, and account detail viewing. This internship gave me a very good opportunity to get exposure to industry-style problem solving and practical implementation using Java programming. It helped me improve my coding skills, logical thinking, debugging ability, and understanding of software development. Overall, it was a great learning experience.

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1 Preface

This internship was an important step in my learning journey and career development. During these 6 weeks, I worked on a mini software project titled “Banking Information System”.

The need for internships is very important because they help students gain practical knowledge beyond classroom learning. This internship helped me understand how real-world software problems can be approached and solved using programming.

My project focused on developing a simple Java-based banking application that performs basic operations such as account creation, login authentication, balance checking, deposit, and withdrawal.

This opportunity provided by Upskill Campus and UCT helped me gain confidence in coding, logical thinking, and software implementation.

During the internship, the work was planned in a structured way:

- Understanding the problem statement

- Designing the solution

- Writing the Java code

- Testing the application

- Preparing the final documentation

Through this internship, I learned not only technical concepts but also patience, problem-solving, and the importance of proper documentation.

I would like to thank Upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd, and everyone who supported me directly or indirectly during this internship.

My message to my juniors and peers is: internships are very useful for skill development, so always try to learn practically and make the best use of such opportunities.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

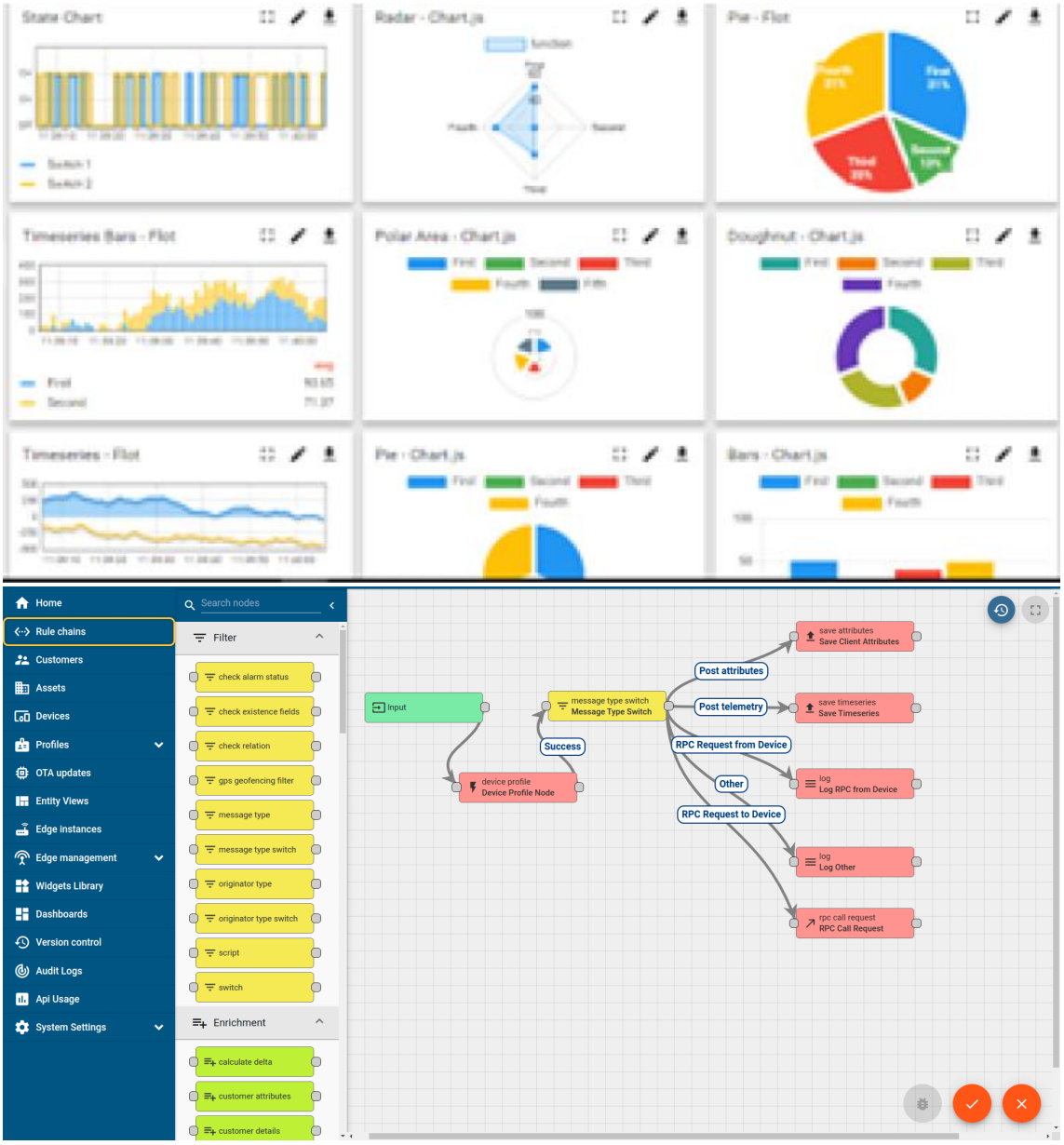
i. UCT IoT Platform (UCT Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

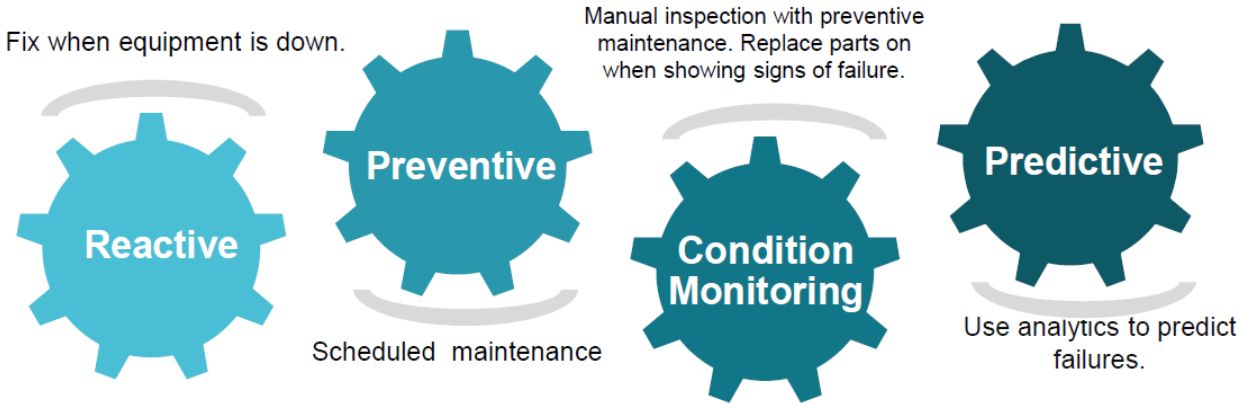
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

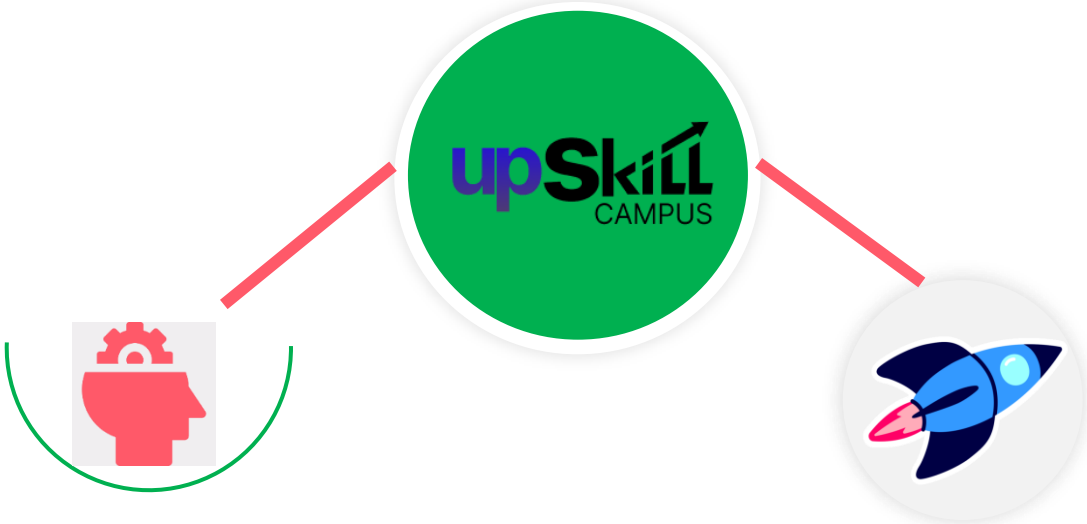
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

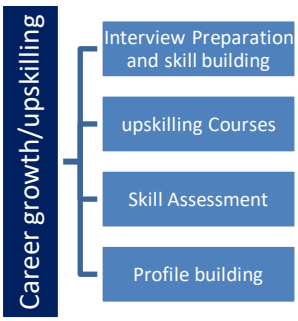
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

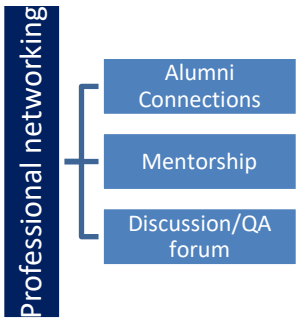


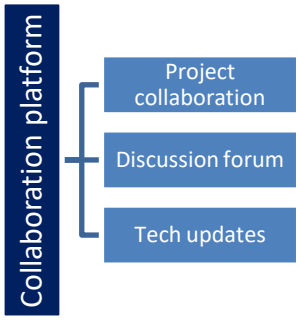
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

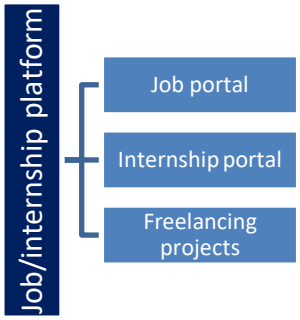
upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/









2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective of this internship project was to design and develop a simple Banking Information System using Java.

The main objectives are:

- To understand the basics of Java application development

- To implement banking operations using programming logic

- To improve problem-solving and coding skills

- To create a menu-driven real-time mini project

- To gain practical exposure through industrial internship experience

2.5 Reference

[1] Java Programming Documentation – Oracle

[2] Upskill Campus Internship Guidelines

[3] Java Scanner Class Reference

2.6 Glossary

Terms	Acronym
BIS	Banking Information System
JAVA	Programming Language used
IDE	Integrated Development Environment
Login	User authentication process
Scanner	Java class used for input

3 Problem Statement

In the assigned problem statement

The problem statement assigned in this internship was to create a simple software-based banking system that can simulate basic banking operations.

The application should allow the user to:

- Create a bank account

- Log in securely using account number and password

- Check account balance

- Deposit money

- Withdraw money

- View account details

- Logout safely

The purpose of this problem statement is to understand how banking-related functions can be implemented in a Java console application using programming fundamentals.

4 Existing and Proposed solution

There are many existing banking applications available today that provide advanced features such as online fund transfer, mobile banking, ATM support, account history, and database integration.

However, such systems are complex and not suitable for beginners who are just learning programming.

Limitations of existing solutions for learning purposes:

- Too complex for beginners

- Requires advanced database and backend integration

- Hard to understand without strong programming knowledge

Proposed Solution:

My proposed solution is to develop a simple Banking Information System using Java as a console-based application.

This system allows users to:

- Create an account

- Login using account number and password

- Check account balance

- Deposit and withdraw money

- View account details

Value Addition:

This project provides a beginner-friendly understanding of how a banking system works and helps in learning Java concepts practically.

Code Submission (GitHub Link):

https://github.com/SravanthiSamurothu/upskillcampus

Report Submission (GitHub Link):

[Paste your GitHub PDF report link here]

4.1 Code submission (Github link)

https://github.com/SravanthiSamurothu/upskillcampus

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

User

↓

Create Account / Login

↓

Banking Menu

↓

Check Balance / Deposit / Withdraw / Account Details

↓

Logout / Exit

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

Start

↓

Display Main Menu

↓

1. Create Account

2. Login

3. Exit

↓

If Login Successful

↓

Display Banking Options

↓

Check Balance / Deposit / Withdraw / Details / Logout

↓

End

5.3 Interfaces (if applicable)

This project is a console-based application, so the interface is text-based.

Main Interface:

- Create Account

- Login

- Exit

User Interface after Login:

- Check Balance

- Deposit Money

- Withdraw Money

- Account Details

- Logout

The interaction happens through keyboard input using the Scanner class in Java.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Result	Status
Create Account	Name, Acc No, Password, Deposit	Account created successfully	Pass
Login Success	Correct Acc No + Password	Login successful	Pass
Login Failure	Wrong password	Invalid login	Pass
Deposit	Valid amount	Balance increases	Pass
Withdraw	Valid amount within balance	Balance decreases	Pass
Withdraw Failure	Amount > balance	Error shown	Pass
Check Balance	Select balance option	Current balance displayed	Pass
Account Details	Select details option	Details displayed	Pass

6.2 Test Procedure

The application was tested manually by entering different user inputs through the console.

Testing included:

- Creating a new account

- Logging in with valid and invalid credentials

- Depositing money

- Withdrawing money with sufficient and insufficient balance

- Checking balance

- Viewing account details

- Logging out and exiting the program

All major functionalities were tested successfully.

6.3 Performance Outcome

The Banking Information System performed successfully for all intended operations.

Observed outcomes:

- The system accepted user input correctly

- Account creation worked properly

- Login authentication worked correctly

- Deposit and withdrawal operations updated the balance correctly

- Invalid operations were handled with proper messages

Since this is a beginner-level console-based Java application, the system performed efficiently without any major delay or memory issue.

7 My learnings

Through this internship project, I learned many important technical and practical concepts.

My key learnings include:

- Writing a complete Java console-based application

- Using Scanner class for user input

- Applying conditional statements and loops

- Using switch-case for menu-driven programming

- Implementing basic banking logic

- Testing and debugging Java code

- Preparing a structured technical report

This internship helped me improve my coding confidence and gave me a better understanding of how real-world software systems are designed and implemented.

8 Future work scope

This project can be improved further in the future by adding more advanced features such as:

- Support for multiple accounts
- Database connectivity using MySQL
- Transaction history
- Fund transfer between accounts
- ATM simulation
- GUI-based application
- Online banking features
- Admin panel for account management

These enhancements can make the project more realistic and industry-ready.